

## DATA LAYER

NASA MAST Archive



1146 TESS Light Curves  
(Sectors 12-13)



1146 Cleaned .npz Arrays

## DETECTION LAYER

BLS Period Search  
(0.3-27 days)



217 Periodic Candidates



Phase Folding  
200-point Arrays

## ML LAYER

34 Feature Extraction



Random Forest  
(2000 trees, 89% accuracy)



4-Class Output + Confidence

## PARAMETER LAYER

Trapezoidal Transit  
Model Fitting



Period, Depth, Duration,  
SNR + Error Bars

## OUTPUT LAYER

predictions.csv  
Visualization Plots  
Validation Report